

Simple Linear Regression

Parameter Estimation via MLE

Firuz Kamalov

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Lecture Outline

Example

Geometric Approach

MLE

A Motivating Example: House Prices

Scenario:

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The Core Question: Can we build a simple model to predict the price of a house based on its size?

A Motivating Example: House Prices

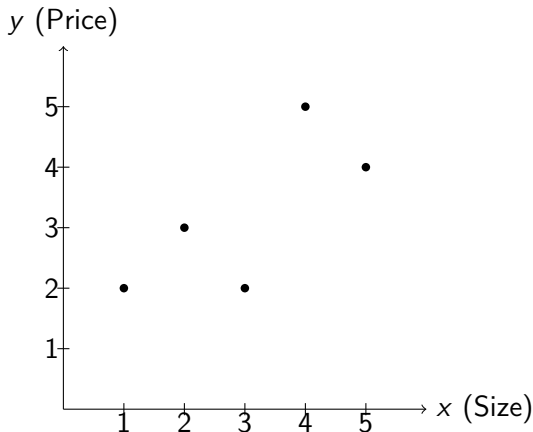
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The Core Question: Can we build a simple model to predict the price of a house based on its size?

Size (sq. m, x)	Price (\$1000s, y)
1	2
2	3
3	2
4	5
5	4

Visualizing the Data: The Scatter Plot

The first step is always to visualize our data. A scatter plot shows the relationship between our two variables.



We observe that as the size increases, the price tends to increase.

Modeling the Relationship: A Straight Line

The simplest way to model this trend is to fit a straight line through the data.

The Model: $y \approx \beta_0 + \beta_1 x$

- ▶ β_0 : The intercept (the price of a 0 sq. m house)
- ▶ β_1 : The slope (the increase in price for each extra sq. m)

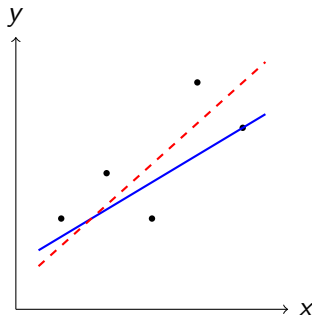
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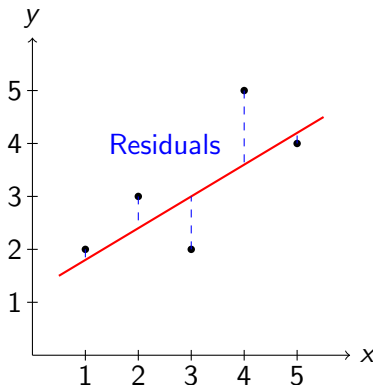
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But there are many lines: which one is the **best**?



Geometric Approach: Minimizing Vertical Distance

We can measure this closeness by the vertical distance from each point to the line. This distance is called the **residual**.



We want to minimize:

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

A More Principled Approach: MLE

Minimizing squared distances is a great geometric intuition. But is there a more formal, statistical justification for it?

Yes! Maximum Likelihood Estimation (MLE).

Instead of just drawing a line, let's propose a probabilistic model.

The Linear Regression Model Assumption: Our data points are not perfectly on the line. The deviation from the line (the error) is random noise, which we assume follows a Normal (Gaussian) distribution.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{where} \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- ▶ The mean of the errors is 0 (the line is correct on average).
- ▶ The errors have a constant variance σ^2 .

MLE for the Straight Line

The assumption $\epsilon_i = y_i - (\beta_0 + \beta_1 x_i) \sim \mathcal{N}(0, \sigma^2)$ is equivalent to saying:

$$y_i | x_i, \beta_0, \beta_1, \sigma^2 \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$$

The probability density of a single data point (x_i, y_i) is:

$$f(y_i | x_i; \beta_0, \beta_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}\right)$$

The Likelihood Function for all n data points is the product of their individual probabilities:

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)$$

Our goal is to find the parameters (β_0, β_1) that **maximize this likelihood**.

Solving for the MLE

As usual, it's easier to maximize the **log-likelihood**, $\ell = \log L$.

$$\ell(\beta_0, \beta_1, \sigma^2) = \log \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right) \right)$$

$$\ell = \sum_{i=1}^n \left[\log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) - \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right]$$

$$\ell = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

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To maximize this function with respect to β_0 and β_1 , we only need to focus on the term that includes them. Maximizing ℓ is equivalent to **minimizing**:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

This is exactly the Sum of Squared Residuals (SSR) from the geometric approach!

Deriving the Optimal Coefficients

Let $S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$. We find the minimum by setting the partial derivatives to zero.

1. **Derivative w.r.t. β_0 :** $\frac{\partial S}{\partial \beta_0} = -2 \sum (y_i - \beta_0 - \beta_1 x_i) = 0$

$$\implies \sum y_i - n\beta_0 - \beta_1 \sum x_i = 0$$

$$\implies n\beta_0 = \sum y_i - \beta_1 \sum x_i \implies \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

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2. **Derivative w.r.t. β_1 :** $\frac{\partial S}{\partial \beta_1} = -2 \sum x_i (y_i - \beta_0 - \beta_1 x_i) = 0$

Solving this system of two equations gives the final formulas for the MLE estimators:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Calculation for Our House Price Data

First, find the means: $\bar{x} = \frac{1+2+3+4+5}{5} = 3$ and $\bar{y} = \frac{2+3+2+5+4}{5} = 3.2$.

Next, we build a table to compute the sums:

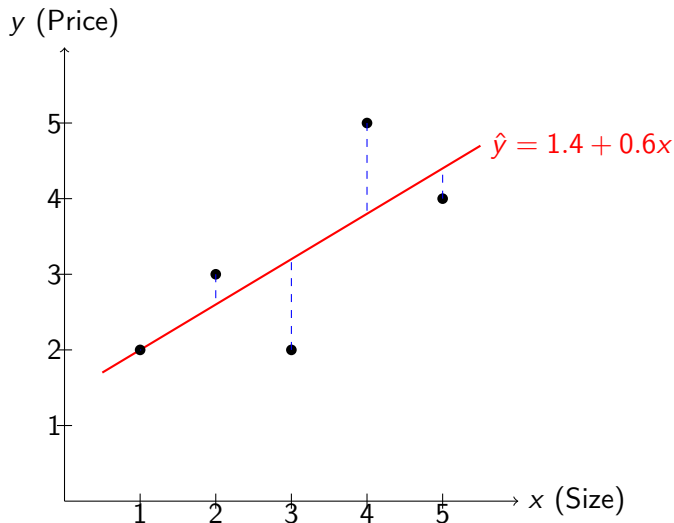
x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	2	-2	-1.2	2.4	4
2	3	-1	-0.2	0.2	1
3	2	0	-1.2	0.0	0
4	5	1	1.8	1.8	1
5	4	2	0.8	1.6	4
Sum				6.0	10

Calculate the slope ($\hat{\beta}_1$): $\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{6.0}{10} = 0.6$

Calculate the intercept ($\hat{\beta}_0$):
 $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3.2 - (0.6)(3) = 3.2 - 1.8 = 1.4$

The Best Fit Line

Plotting our calculated line, $\hat{y} = 1.4 + 0.6x$, on the scatter plot.



Why Linear Regression is Still Important

Advantages

- ▶ **Interpretability:** Very easy to understand and explain.
- ▶ **Simplicity & Speed:** It is computationally inexpensive and fast to train.
- ▶ **Foundation:** It is the basis for many more advanced statistical and machine learning models (e.g., Logistic Regression, GLMs, XAI).

Applications in AI and Data Science

- ▶ **Baseline Model:** It provides a baseline performance that more complex models (like neural networks) must beat.
- ▶ **Forecasting:** Used in economics and finance to predict variables like GDP, sales, or demand based on time or other factors.
- ▶ **Understanding Relationships:** Used in social sciences and

- ▶ **Simple Linear Regression** models the relationship between two variables, a predictor (x) and a response (y), with a straight line.
- ▶ The goal is to find the "best" line, meaning the best values for the intercept (β_0) and slope (β_1).
- ▶ The geometric approach (**OLS**) finds the best line by minimizing the sum of the squared vertical distances (residuals) from the points to the line.
- ▶ **MLE** provides a powerful statistical foundation for this. Assumes that the errors around the regression line are normally distributed. MLE becomes equivalent to OLS.